

Multiple Choice (10 marks)

1. What is the rank of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
2. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
3. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance
4. Statement 1| Layer Normalization is used in the original ResNet paper, not Batch Normalization. Statement 2| DCGANs use self-attention to stabilize training.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.

True / False (10 marks)

Answer True or False

6. True or False: The original ResNets were primarily optimized using the SGD optimizer, leading to their notable performance improvements.
7. True or False: An excessively large step size during training can cause the training loss to increase over epochs, indicating instability in learning.
8. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as the variance of the Maximum Likelihood Estimate (MLE).
9. True or False: Lasso regression is more appropriate for feature selection due to its ability to perform variable shrinkage and selection.
10. True or False: The number of test examples needed decreases if the error rate is larger, as fewer examples suffice for significance.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to convert a list of multiple integers into a single integer.
12. Write a function to check if a string represents an integer or not.

End of Paper